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DEPARTMENT OF BUSINESS MANAGEMENT & ENTREPRENEURSHIP

BACHELOR OF COMPUTER APPLICATIONS

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MINOR PROJECT ON

"EXPLORATORY DATA ANALYSIS"

SUBMITTED BY

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(Bachelor of Computer Applications)

SEMESTER-V

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CERTIFICATE

This is to certify that the work contained in this Project report entitled "**Diwali Sales Analysis**" by **SHIVAM SHUKLA**, **23060100010070** is a faithful record of work that has been carried out by the student, under my/our supervision and the level of work is good for submission. To the best of my/our knowledge, this work has not been submitted for award of any degree or diploma to this University or elsewhere.

Supervisor : .....

ACKNOWLEDGEMENT

There are many people, I would like to thank for their help and support in writing this thesis. First of all, I would like to express special thanks to almighty "GOD" for his blessings. I would like to thanks my respected parents for their guidance, care, constant encouragement, blessings and cooperation given by them. I express the deep sense of gratitude to all my family members.

I feel extremely privileged to express my deep sense of gratitude and regard to my supervisor

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At last I would like to thank all those who could not find a separate mention but have helped me directly or indirectly.

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Abstract

The project titled **"Diwali Sales Analysis Using Python and Excel"** focuses on analyzing customer purchasing behavior during the Diwali festival season. Diwali is one of the largest shopping festivals in India, during which businesses experience a massive increase in sales across multiple product categories. However, raw sales data alone does not provide meaningful business insights. Without proper analysis, businesses cannot identify high-performing product segments or profitable customer groups.

This project applies **Exploratory Data Analysis (EDA)** techniques to study Diwali sales data systematically. The dataset consists of more than 11,000 sales records containing information such as customer ID, gender, age group, marital status, state, occupation, product category, number of orders, and purchase amount.

To perform this analysis, tools such as:

- Microsoft Excel (for data overview and initial inspection)
- Python Programming Language
- NumPy (for numerical operations)
- Pandas (for data manipulation)
- Matplotlib & Seaborn (for visualization)
- Jupyter Notebook (for interactive coding environment)

The project begins with data cleaning, including removal of null values and unnecessary columns. After preprocessing, statistical summaries and grouping operations were performed. Various visualizations such as bar charts and count plots were created to understand patterns in sales behavior.

The analysis revealed that:

- Female customers contributed a higher portion of total revenue.
- Age group 26–35 showed maximum purchasing activity.
- Certain states generated significantly higher revenue.
- Specific product categories dominated festive sales.

This project demonstrates how data analytics transforms raw data into strategic insights that can improve business decision-making, marketing strategies, and inventory management during festive seasons.

Introduction

Festivals play a crucial role in boosting economic activity in India. Among them, Diwali is considered the biggest shopping festival. During this time, consumers purchase a wide range of products such as electronics, clothing, home décor, gifts, and automobiles. As a result, companies generate large volumes of sales data.

However, collecting data is not sufficient. The true value lies in analyzing this data to extract meaningful information. Many organizations face challenges such as:

- Identifying their target customers
- Understanding purchasing patterns
- Improving marketing effectiveness
- Optimizing stock levels
- Predicting demand trends

To overcome these challenges, this project applies data analysis techniques to understand Diwali sales trends. The project follows a systematic approach starting from raw data collection to final insight generation.

The dataset used in this project includes demographic information such as age, gender, occupation, state, and marital status, along with transaction details like product category, number of orders, and purchase amount.

By performing Exploratory Data Analysis (EDA), the project identifies:

- Revenue distribution by gender
- Age group contribution to sales
- Top-performing states
- Occupation-based purchasing behavior
- Popular product categories

The findings of this project can help businesses design targeted marketing campaigns, manage inventory efficiently, and increase festive season profit

Objective

The primary objective of this project is to analyze Diwali sales data and extract meaningful business insights using data analytics tools.

The specific objectives are:

1. To analyze customer demographics such as gender, age group, and marital status.
2. To determine which age group contributes the highest revenue.
3. To identify gender-based purchasing differences.
4. To perform state-wise sales analysis.
5. To analyze occupation-wise spending patterns.
6. To identify high-performing product categories.
7. To apply data cleaning and preprocessing techniques.
8. To use visualization tools for better understanding of patterns.
9. To demonstrate practical implementation of Python libraries such as Pandas and NumPy.
10. To provide recommendations based on analytical findings.

This project aims not only to analyze historical data but also to build a foundation for future predictive modeling.

Proposed Model

The proposed model follows a structured data analytics pipeline consisting of the following phases:

Phase 1: Data Collection

Sales data is collected in CSV format containing 11,000+ records.

Phase 2: Data Cleaning

- Removal of null values
- Dropping unnecessary columns such as “Status”
- Converting data types
- Renaming columns for clarity

Phase 3: Data Preprocessing

- Checking descriptive statistics
- Grouping data using Pandas functions
- Aggregation using sum() and count()
- Filtering relevant features

Phase 4: Data Analysis

- Gender-based sales grouping
- Age group analysis
- State-wise revenue comparison
- Occupation-based spending analysis
- Product category performance evaluation

Phase 5: Data Visualization

- Bar charts
- Count plots
- Sales comparison graphs

Phase 6: Insight Generation

Findings are interpreted to support business strategy decisions.

This structured pipeline ensures systematic data handling and accurate results.

Feasibility Analysis

Feasibility analysis determines whether the project can be successfully implemented.

Technical Feasibility

The project uses Python, an open-source programming language. Libraries such as Pandas and NumPy allow efficient data manipulation. Jupyter Notebook provides an interactive development environment.

Economic Feasibility

All tools used are free and open-source. No additional financial investment is required.

Operational Feasibility

The project can be easily implemented in real business environments. The analysis process is simple and replicable.

Time Feasibility

The project was completed within the academic semester schedule.

Thus, the project is technically, economically, operationally, and temporally feasible

Requirement Analysis

Hardware Requirements:

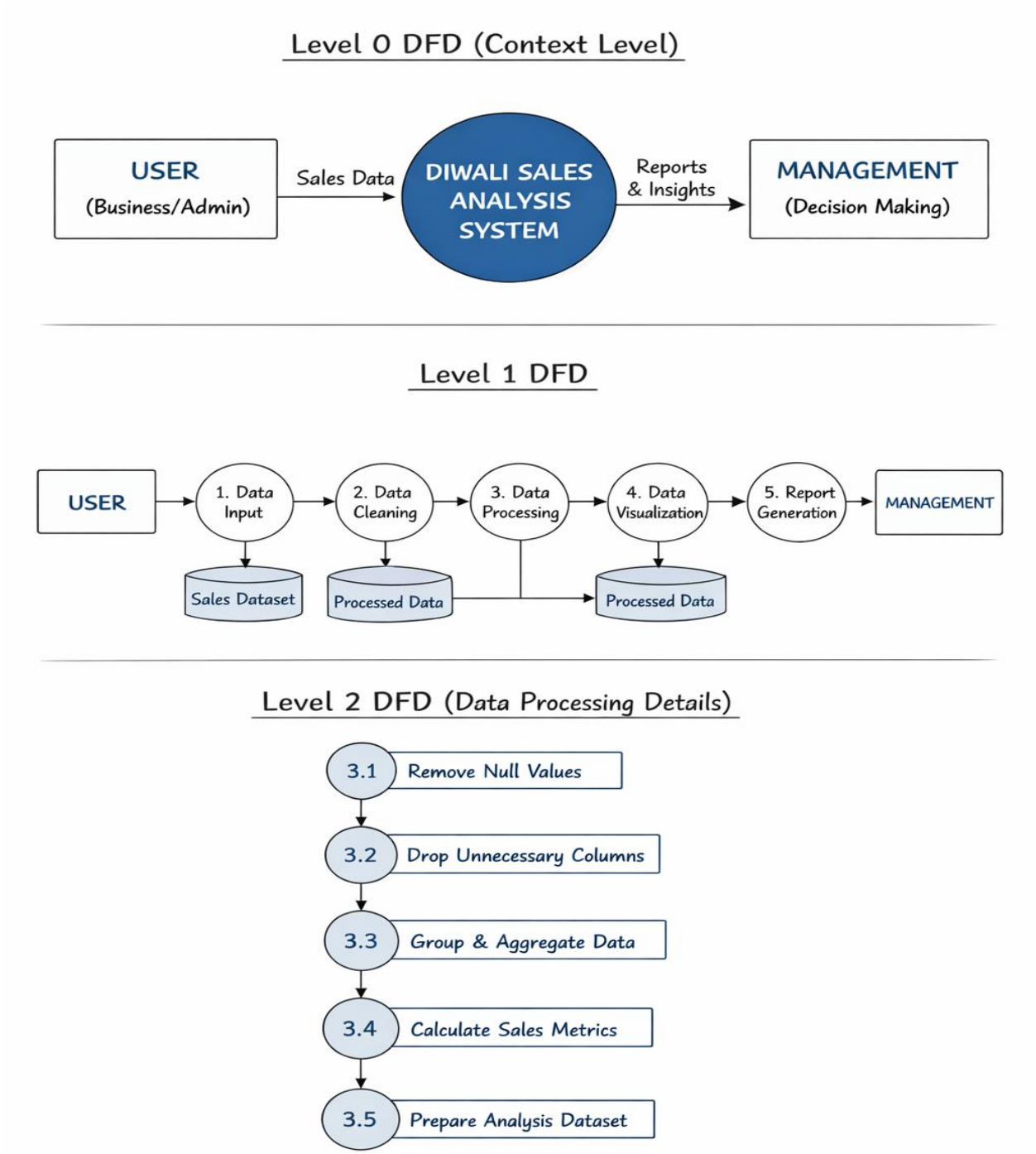
- Processor: Intel i3 or higher
- RAM: Minimum 4GB
- Storage: 500GB hard disk
- Monitor, Keyboard, Mouse

Software Requirements:

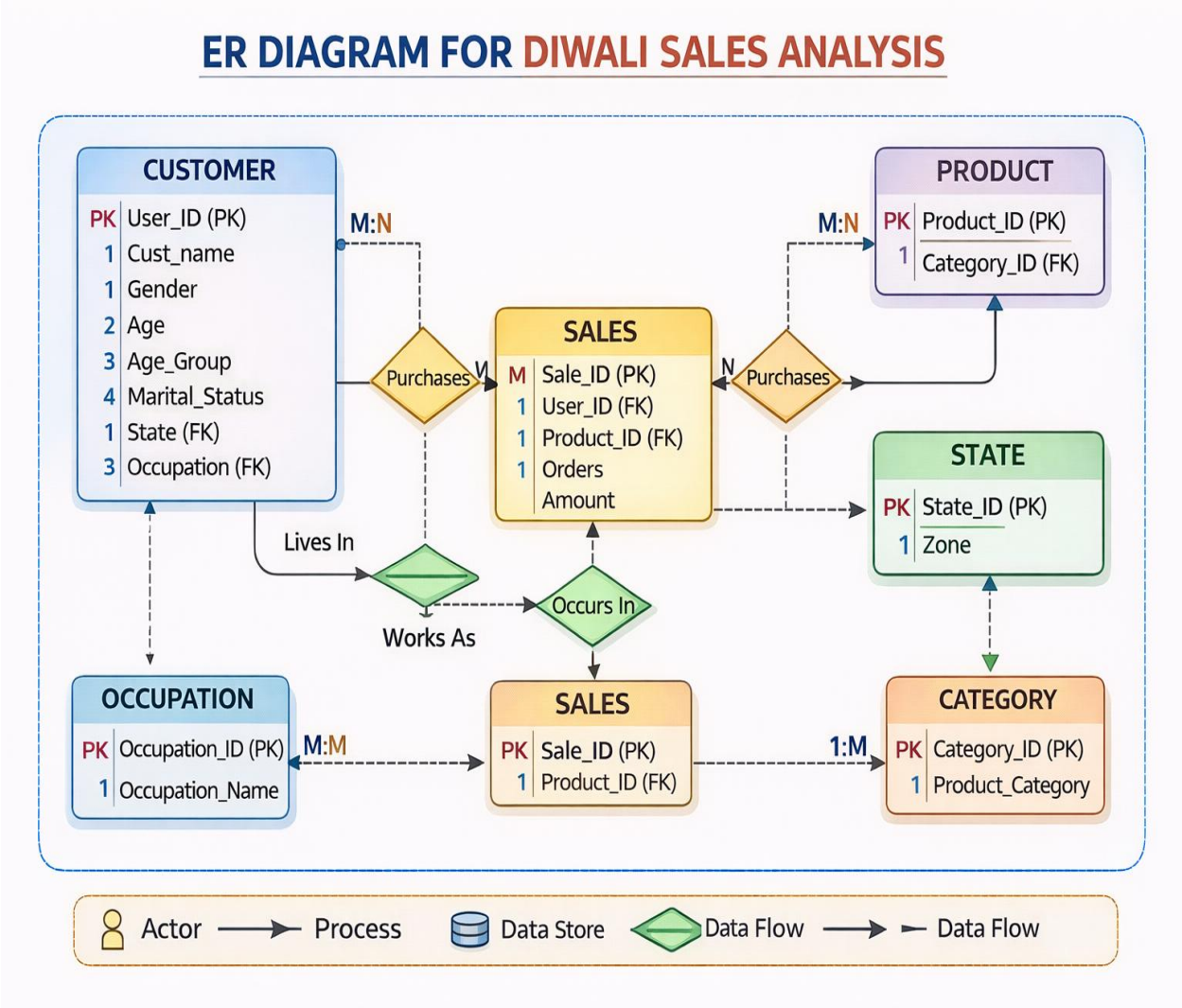
- Operating System: Windows 10/11
- Python 3.x
- Jupyter Notebook
- Microsoft Excel
- Libraries:
    - NumPy
    - Pandas
    - Matplotlib
    - Seaborn

These requirements are sufficient to perform data cleaning, analysis, and visualization smoothly.

Data Flow Diagrams



ER Diagram


ER DIAGRAM FOR DIWALI SALES ANALYSIS
CUSTOMER
PK User_ID (PK)
1 Cust_name
1 Gender
2 Age
3 Age_Group
4 Marital_Status
1 State (FK)
3 Occupation (FK)
M:N
Purchases
M
SALES
M Sale_ID (PK)
1 User_ID (FK)
1 Product_ID (FK)
1 Orders
Amount
N
Purchases
M:N
PRODUCT
PK Product_ID (PK)
1 Category_ID (FK)
STATE
PK State_ID (PK)
1 Zone
Lives In
Works As
Occurs In
OCCUPATION
PK Occupation_ID (PK)
1 Occupation_Name
M:M
SALES
PK Sale_ID (PK)
1 Product_ID (FK)
1:M
CATEGORY
PK Category_ID (PK)
1 Product_Category
Actor
Process
Data Store
Data Flow
Data Flow

Future Work

The project can be extended in the following ways:

1. Multi-year Diwali sales comparison for trend analysis.
2. Application of Machine Learning algorithms for sales prediction.
3. Creation of interactive dashboards using Power BI or Tableau.
4. Development of web-based analytics application.
5. Integration of customer feedback analysis.
6. Predictive inventory management.

These improvements can make the project more advanced and industry-ready.

Conclusion

The Diwali Sales Analysis project successfully converted raw festive season sales data into meaningful insights. The study revealed that female customers and the 26–35 age group contribute significantly to overall revenue.

State-wise and occupation-wise analysis helped identify high-value customer segments. Product category analysis provided insights into top-selling items during the festive season.

The project demonstrates the importance of data analytics in modern business environments. By using Python libraries and data visualization techniques, organizations can make informed decisions, increase profitability, and enhance customer satisfaction.

Overall, this project highlights how data-driven strategies can improve festive sales performance and business growth.